EXTENDING AN INTEGRAL MORPHISM OVER A QUASI-COMPACT OPEN IMMERSION

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 2009-2021, immediately following the statement at 2001-2007.

Editorial note. The 'follows formally' phrase at source line 2017 is retained verbatim; the uniqueness reading recorded in proof_roadmap is the reviewer's checklist, not author text.

Editorial note. Three cross-chapter pointers (Morphisms of Spaces) are recorded in assembly.external_references.

Allowed standard prerequisites (editorial).

- Normalization of an algebraic space in another. Morphisms of Spaces, Section spaces-morphisms-section-normalization-X-in-Y, cited at proof lines 2012-2013. Constructs nu : Y' -> Y, the normalization of Y in Z.
- Normalizations are integral. Morphisms of Spaces, Lemma spaces-morphisms-lemma-characterize-normalization, cited at proof lines 2014-2016. nu is integral by the characterization of normalizations.
- Normalization restricts to the source over the open and agrees with integral extensions. Morphisms of Spaces, Lemmas spaces-morphisms-lemma-properties-normalization and spaces-morphisms-lemma-normalization-in-integral, cited at proof lines 2017-2021. Give the final statement: over V, Y' recovers Z (Z is V-isomorphic to the inverse image of V in Y').

Source reference. Original reference 0823: focused-seeds/geometry/S9-decent-extend-integral-morphism/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0825: focused-seeds/geometry/S9-decent-extend-integral-morphism/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0ABP: focused-seeds/geometry/S9-decent-extend-integral-morphism/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:decent-spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 0BAZ: focused-seeds/geometry/S9-decent-extend-integral-morphism/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let $j : V \to Y$ be a quasi-compact open immersion of algebraic spaces over S. Let $\pi : Z \to V$ be an integral morphism. Then there exists an integral morphism $\nu : Y' \to Y$ such that Z is V-isomorphic to the inverse image of V in Y'.*

Proof. Since both j and π are quasi-compact and separated, so is $j \circ \pi$. Let $\nu : Y' \to Y$ be the normalization of Y in Z, see Morphisms of Spaces, Section 0BAZ. Of course ν is integral, see Morphisms of Spaces, Lemma 0823. The final statement follows formally from Morphisms of Spaces, Lemmas 0ABP and 0825. □